Beyond the UMR Parsing Shared-Task Rankings

A Closer Look at Czech Results

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September 26, 2026

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unless otherwise stated

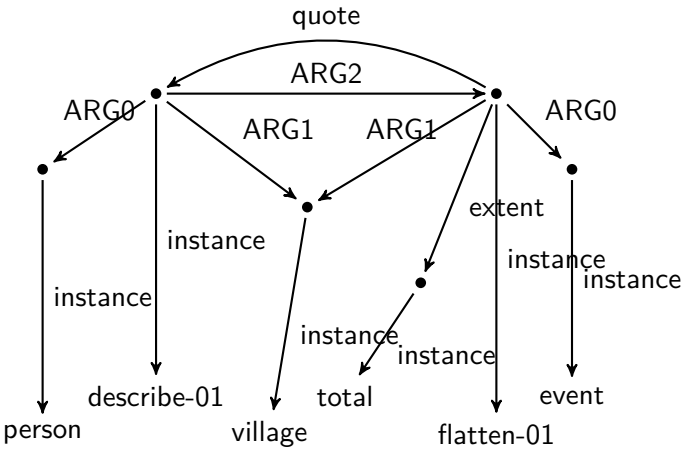
Uniform Meaning Representation

- Sentence level graph
- Document level relations
- Node-token alignment
- (Morphemic segmentation and glosses)

UMR: Sentence Level Graph

She described the village as totally flattened.

```
(s13d / describe-01
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     :refer-person 3rd
     :refer-number singular)
  :ARG1 (s13v / village)
  :ARG2 (s13f / flatten-01
     :ARG0 (s13e / event)
     :ARG1 s13v
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     :quote s13d)
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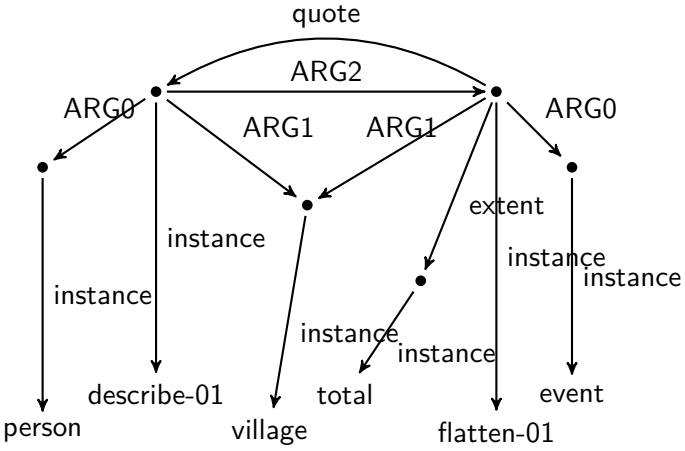


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      :
  :a
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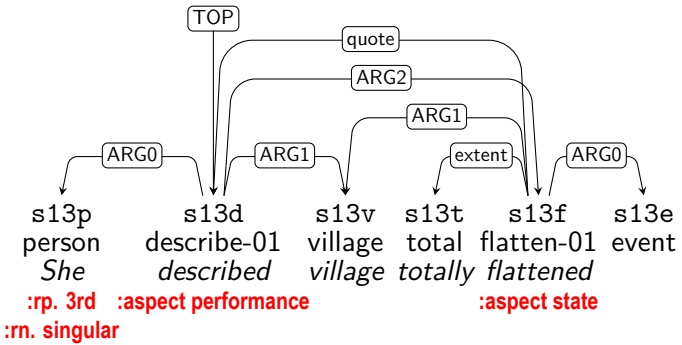
PENMAN notation



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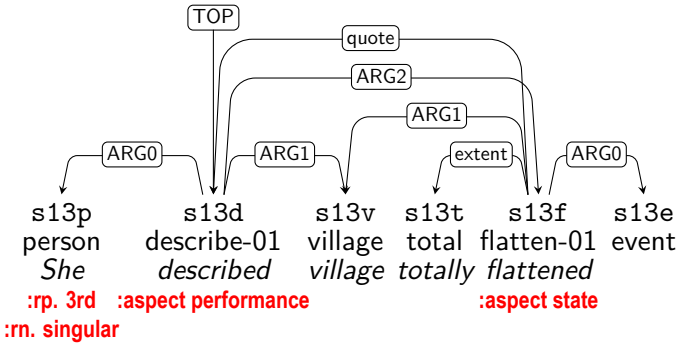
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UMR: Alignment (Nodes to Tokens)

1	2	3	4	5	6	7	8
She	*described*	*the*	*village*	*as*	*totally*	*flattened*	*.*

```
s13p: 1-1
s13d: 2-2
s13v: 4-4
s13t: 6-6
s13f: 7-7
s13e: 0-0
```

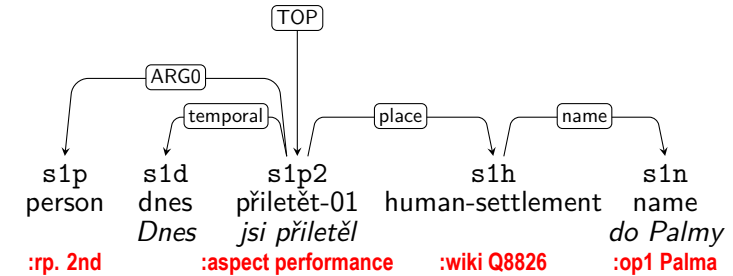


UMR: Discontinuous Alignment

1	2	3	4	5	6
Dnes	*jsi*	*do*	*Palmy*	*přiletěl*	.
Today	you have	to	Palma	flown in	.

'You arrived in Palma today.'

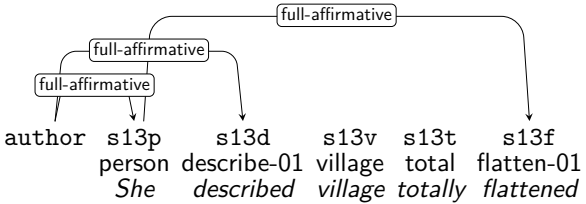
```
s1p: 0-0
s1d: 1-1
s1p2: 2-2,5-5
s1h: 0-0
s1n: 3-4
```



UMR: Document Level Relations

- *Leyte Governor Rosette Lerias said the latest census indicated there were 375 houses with 3,000 people living in Guinsaugon before the landslide.*
- *She described the village as totally flattened.*

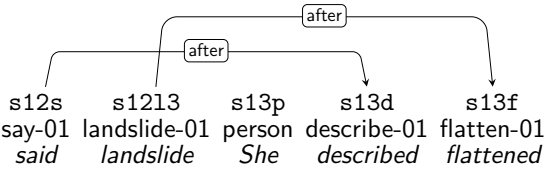
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        (author :full-affirmative s13p)
        (s13p :full-affirmative s13f))
    :temporal ((s12s :after s13d)
        (s12l3 :after s13f))
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        (s12p :same-entity s13p)
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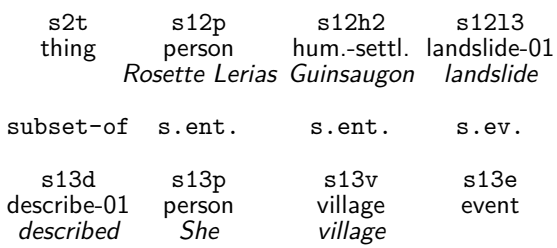
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Uniform Meaning Representation release 2.1

- Arapaho ... cleaned (using our validating script)
- Chinese ... new data annotated for the shared task
- Czech ... new data annotated for the shared task
- English ... cleaned, removed duplicate document (and overlooked another)
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Parallel Universal Dependencies (subset 100 sentences)

- Czech
- English
- Italian

Data Divergences

- Manually annotated vs. automatically converted (from AMR, PDT…)
  - Missing document-level relations
    - Or partial: PDT has coreference but not modal or temporal
  - Missing or extra relations even on sentence level
  - Missing alignments
- Incomplete manual annotation
  - Missing document-level relations
  - Missing alignments
  - Missing `:wiki`
- Stage 0 vs. Stage 1 annotation
  - Stage 0: `:actor`, `:undergoer`, `:recipient`, `:experiencer`, `:stimulus`, `:theme`
  - Stage 1: `:ARG0` – `:ARG5`
  - Modal annotation: Stage 0 in sentence, Stage 1 in document

- **Dirty vs. clean training data (test data is clean)**
- Do not penalize system if it predicts something the test data lacks

Data Sizes

	Training data						Test data		
	Clean			Dirty					
	Files	Sents.	Tokens	Files	Sents.	Tokens	Files	Sents.	Tokens
Arapaho	1	53	274	2	292	1,627	2	55	274
Chinese	20	557	15,375	1	1,435	18,656	15	236	6,467
Czech	**6**	**103**	**1,587**	**6,516**	**159,906**	**2,537,317**	**5**	**220**	**4,048**
English	4	180	2,023	570	29,872	292,023	5	195	4,092
Italian	—	—	—	—	—	—	1	100	2,212
Latin	—	—	—	3	1,049	19,139	1	50	889
Navajo	1	5	60	3	337	2,663	1	163	1,194

UMR release 2.2 (includes system outputs): `http://hdl.handle.net/11234/1-6132`

Evaluation Metric

- Decompose UMR into triples
  - node `x` – `:relation` – node `y`
    - even document-level relations (add a few special nodes)
  - node `x` – `:attribute` – value
    - (optionally also: node `x` – is *top node* – yes)
  - node `x` – `:instance` – concept
- Compute F_1 of gold vs. system triples
- Problem: How to map nodes between gold and system graphs?

Evaluation Metric

- Encourage systems to predict alignment correctly
- Use alignment to find mapping between nodes
  - Unlike, e.g., *smatch*
- Already implemented for inter-annotator comparison: ***ju:mætʃ***

⇒ If you mess up alignment, you **FAIL** badly!

Systems Participated in the Shared Task

ORANGE, Johannes Heinecke – Munshi Asadullah (2026):

- data: raw text, all training data used
- sentence-level: three T5-family models
  the best model per language was selected ⇒ for Czech, monolingual mT5 model
- document-level: a separate Qwen3 8B model
- alignment: a heuristic lemma-to-concept matching strategy

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PDT-to-UMR, Štěpánek et al. (2025), Lopatková et al. (2025):

- data: Czech (and Latin) deep syntactic graphs, manually annotated
- rule-based conversion from the PDT corpus ⇒ **Czech dirty training data**
- document-level: only coreference

Results Computed for the *Czech Shared-Task Test Set*

	Shared Task			
	Czech		Average (7 languages)	
	ORANGE	SEMA	ORANGE	SEMA
Overall *ju:mætʃ*	**15.87**	**26.52**	**19.35**	**19.43**

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F_1 on mapped nodes	52.10	48.72	48.82	39.81

Main observations:

- almost the same, **rather low**, macro-average across 7 language, ORANGE much lower for Czech
- the main reason might lie in suboptimal node–token alignment
- performance on mapped nodes much better

New Experiment

Idea:

- manual correction of the node–token alignment predicted by the two systems
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- Czech PDT test set: 4 files with 120 sentences (2,194 tokens)
  - matches the clean training data
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Systems:

- ORANGE
- SEMA
- PDT-to-UMR automatic conversion

Results with Corrected Alignment (for the *Czech PDT Test Set*)

	ORANGE		SEMA		PDT-to-UMR
	original	corrected	original	corrected	
Overall *juːmæʧ*	16.85	**48.59**	25.73	**35.57**	**51.38**

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Proportion of mapped nodes	40.62	81.53	51.28	65.44	80.70

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F_1 on mapped nodes	56.01	60.52	50.38	53.71	62.16

Main observations:

- manual alignment correction resulted in substantially better results
- much stronger impact on ORANGE
  $\Rightarrow$ ORANGE is the **winner** after manual alignment correction

Qualitative Analysis of the Outputs with Corrected Alignment

Data:

- 1 file from the Czech PDT test set:
  28 sentences, 550 tokens covered by 439 gold nodes
  journalistic text

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	ORANGE		SEMA		PDT-to-UMR	
	F_1*	*ju:mætʃ*	F_1*	*ju:mætʃ*	F_1*	*ju:mætʃ*
overall score	59.94 (↓)	**50.10 (↑)**	55.02 (↑)	**31.11 (↓)**	61.55 (↓)	**53.02 (↑)**

* F_1 on mapped nodes (triples governed by unmapped nodes are ignored);
Symbols ↑ and ↓ indicate performance for the analyzed file with respect to the Czech PDT test set with corrected alignment.

ORANGE: Qualitative Analysis

"Coverage":

- outputs cover entire sentences (with only negligible omissions)

Sentence-level graph:

- outputs nicely mirror the dirty training data ⇒ **easy to interpret**
  - ORANGE vs. PDT-to-UMR: *ju:mæʧ* = 81.91 %, F_1* = 88.08 %
- concepts and relations: rather (deep) syntax than semantics
  - lemmata as concepts → limited abstraction
  - very limited use of Named Entities and their UMR-specific representation
- attribute: reasonable quality (modal-strength as the only exception)

Document-level graph:

- NO document-level outputs (modal, temporal, coreference)

SEMA: Qualitative Analysis

"Coverage":

- outputs often do not cover the entire processed sentence
- the system tends to hallucinate

Sentence-level graph:

- overall structure is often inadequate $\Rightarrow$ **difficult to interpret**
- attempts to produce UMR-specific representation
  - problems with lemmatization
  - deverbal nouns $\rightarrow$ verbs (BUT often a source of structural issues)
  - structures involving abstract concepts (BUT recursively embedded now and then)
- attribute: comparable quality, outputs include modal-strength

Document-level graph:

- rarely includes modal annotation (NO modal, NO temporal structure)

Performance in More Detail (with Corrected Alignment, 1 file)

	ORANGE		SEMA		PDT-to-UMR	
	F_1*	*ju:mætʃ*	F_1*	*ju:mætʃ*	F_1*	*ju:mætʃ*
overall score	59.94	50.10	55.02	31.11	61.55	53.02
concept	69.63	57.58	62.06	33.69	69.63	57.04
aspect	57.14	51.46	69.39	44.93	57.52	51.16
modal-strength	–	–	62.50	40.18	–	–
polarity	94.74	94.74	57.14	47.06	94.12	94.12
refer-number	72.58	68.18	66.67	37.50	76.98	73.12
refer-person	54.55	46.15	42.11	21.05	53.33	47.06

Summary

Summary from the Shared Task:

- very complex task, unsatisfactory results
- add alignment guidelines to UMR guidelines?

https://ufal.mff.cuni.cz/umr-parsing

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New findings:

- node–token alignment as the main bottleneck in the rankings
- scores improved substantially when corrected (Czech)
- ORANGE as the Shared Task winner ⇒ promising pre-processing tool

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